ARAB YOUTH SPACE HACKATHON 2026 · CHALLENGE 813 · FINAL

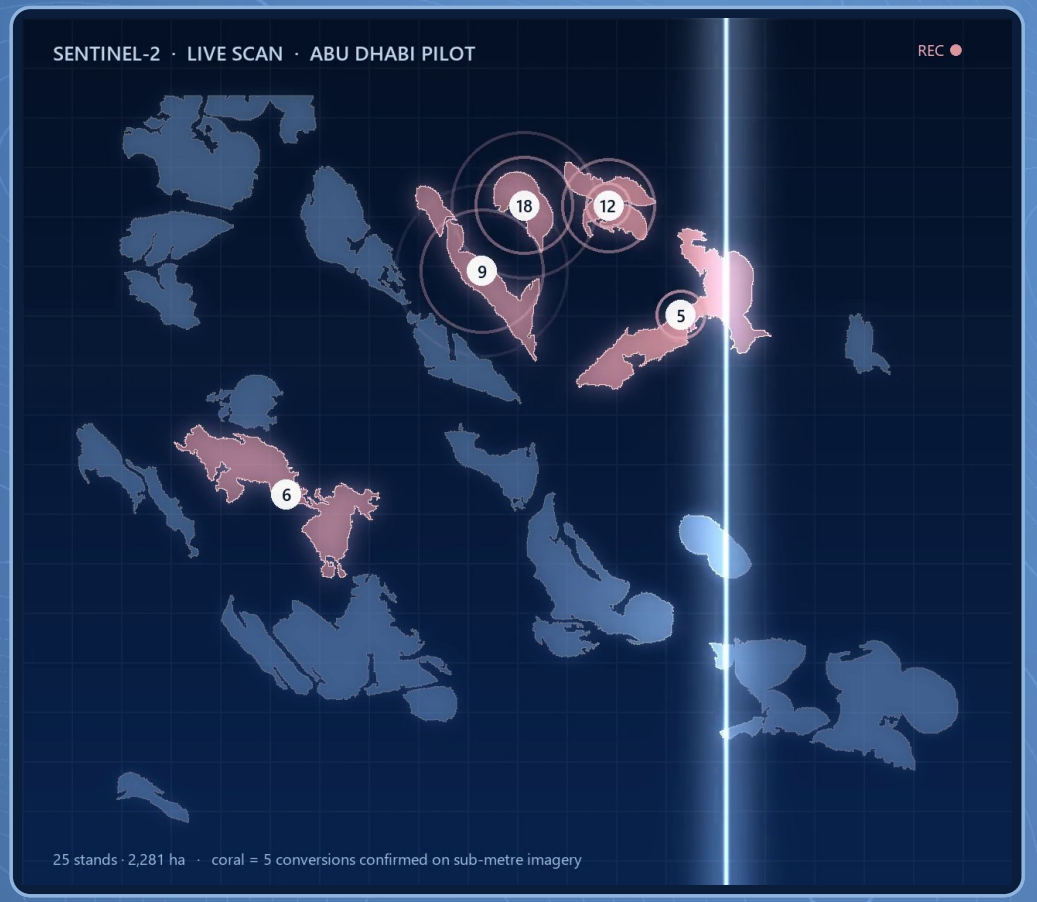
Blue Carbon Guardian

Tide-aware satellite monitoring for Gulf mangroves, from alert to evidence.

Team: Blue Athar · الأثر الأزرق

Theme: Ecosystem Health, Biodiversity & Blue Carbon

91	5	73 %	≈ 1
stands monitored	conversions confirmed	simulated partial loss caught (chance 13 %)	false alarm per stand-year



Study area: Abu Dhabi, UAE

Built in Bahrain

Nahla Nabil · Based in: Bahrain

THE TEAM

A solo project: one engineer, every role



Nahla Nabil

Solo founder · AI engineer

Bahrain

linkedin.com/in/nahla-nabil-876597211

EDUCATION

BSc Artificial Intelligence · Arab Open University, Bahrain, UK · final semester · First-Class Honours standing

RESEARCH

- 2 IEEE papers: ICCA 2025 and ICAISET 2026
- **3rd paper, presenting 29 Oct 2026: PIGNN-UQ, uncertainty-aware blue carbon crediting**

- IEEE and ACM member

EXPERIENCE

- IT intern · Ministry of Interior, Bahrain
- AI & LLM engineering · Doo Bahrain, FlyRank US
- Data Analysis

COMMUNITY

NASA Space Apps Challenge Local Lead, Manama 2026

ROLES & RESPONSIBILITIES IN THIS PROJECT

Project lead & owner

problem framing, user needs, scope, every claim

EO data engineer

778 Sentinel-2 dates, EnMAP reader, sub-metre checks

Modelling & statistics

tide-aware model, stand + cell alert, calibration

Validation & integrity

real events, simulations, unseen stands

Product & delivery

dashboard, website, site reports, notebook

Business & impact

customers, pricing hypotheses, carbon ranges

Incubation gap we will close: a mangrove ecologist mentor · a restoration or agency partner · a GIS analyst

THE PROBLEM

Restoration is scaling faster than monitoring

100 M	**UAE mangrove pledge by 2030** COP26, 2021 · Abu Dhabi Mangrove Initiative
100 M+	**Saudi mangrove pledge by 2030** Saudi Green Initiative
×4	**Bahrain mangrove pledge by 2035** COP26, 2021 · national net-zero path
Feb 2026	**first GCC mangrove monitoring guide** EAD + IUCN Mangrove Specialist Group

TODAY field surveys and one-off studies, no standing alert.
WE SOLVE a calibrated alert every ~5 days that says where to inspect and what carbon is at stake.

THIS IS HAPPENING NOW · STAND 5, ABU DHABI



Sentinel-2 true colour (contains modified Copernicus Sentinel data). Our alert began on 25 Oct 2022, during the works.

Sources: UAE 100 M by 2030 — MOCCAE, COP26 (2021) · Saudi 100 M+ — Saudi Green Initiative · Bahrain ×4 by 2035 — COP26 (2021) · GCC mangrove monitoring guide — Environment Agency – Abu Dhabi with IUCN Mangrove Specialist Group (Feb 2026) · Guidelines for Mangrove Restoration in the UAE — EAD, MOCCAE, Emirates Nature–WWF

LIVE PRODUCT · ABU DHABI PILOT

The monitor

25 mangrove stands (2,281 ha), Sentinel-2 from 2020 to October 2026. Click a stand on the map or in the table. Switch to **200 m cells** to see where inside a stand the change is.

Stands | 200 m cells

Colour = condition, last 12 months vs 2020–21



1 · Every stand coloured by its condition

Stand 5

severe decline

No active alert. 3 alert episodes 2022–26 (combined stand + cell rule).

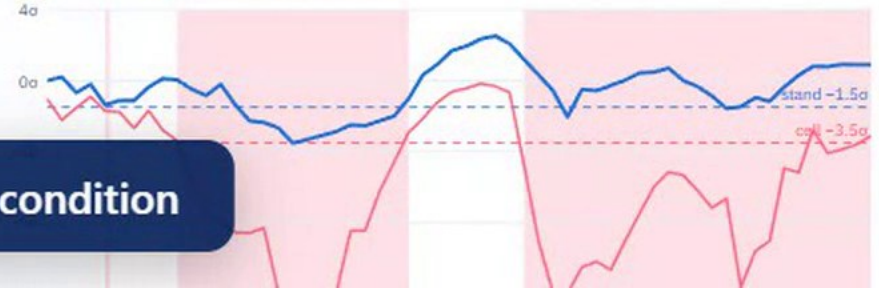
176.5 ha
stand area

19.1 kt C
carbon stock (13.9–24.4)

-111 %
NDVI, last 12 months vs 2020–21

27 of 45 cells
in decline · 72.2 ha to inspect

ALERT SCORE, 2022–2026



BUSINESS USE CASE & END USERS

Who uses it, and for which decision

Growth and protection, together: the UAE is planting 100 million mangroves as it builds its coast. This monitor helps both succeed.

Environment agencies

EAD, MOCCAE, Saudi NCVC

THEIR DECISION

Which stands do we inspect this month?

Today: periodic field surveys

monthly alert + 200 m cell map: inspect only flagged cells

Planting & carbon developers

ADNOC, Red Sea Global, carbon projects

THEIR DECISION

Is the carbon we report still there?

Today: plots and drone campaigns

condition + carbon stock with a range, a site report

Developers & regulators

EIA consultants, permit offices

THEIR DECISION

Did works stay outside the mangrove line?

Today: comparing two images by eye

an alert when works enter mangrove cells, dated evidence

New scene — every ~5 days → **Alert** — ~1 false alarm / stand-yr → **Cell map** — where to look → **Inspection** — only flagged cells → **Site report** — carbon at stake

Five independent data sources, six years, 91 stands



Pilot: 25 stands · 2,281 ha (WorldCover 2021 patches ≥ 3 ha). +66 unseen stands in Abu Dhabi N/W and Saudi Tarut Bay test transfer.

Sentinel-2 L2A
ESA / Copernicus · 10–20 m time series
778 dates

EnMAP hyperspectral
DLR · 224 bands · independent check
2 scenes

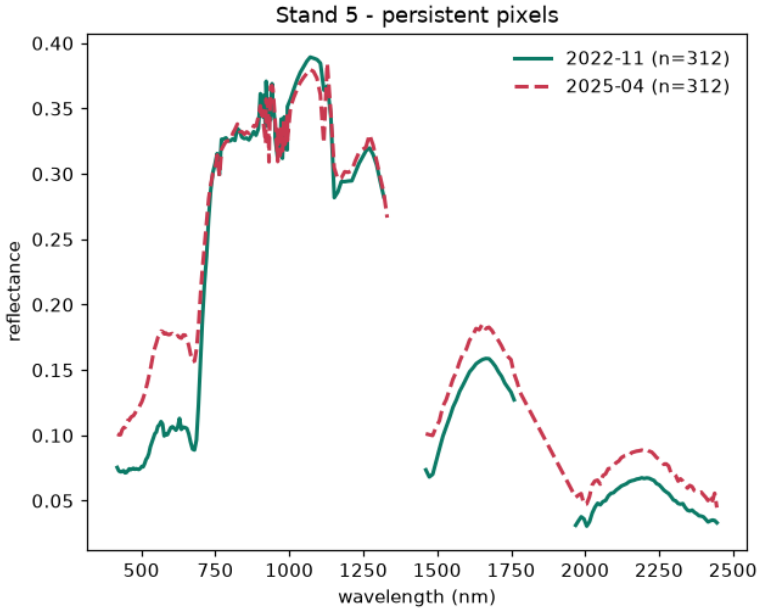
ESA WorldCover 2021
10 m land cover · stand outlines
CC BY 4.0

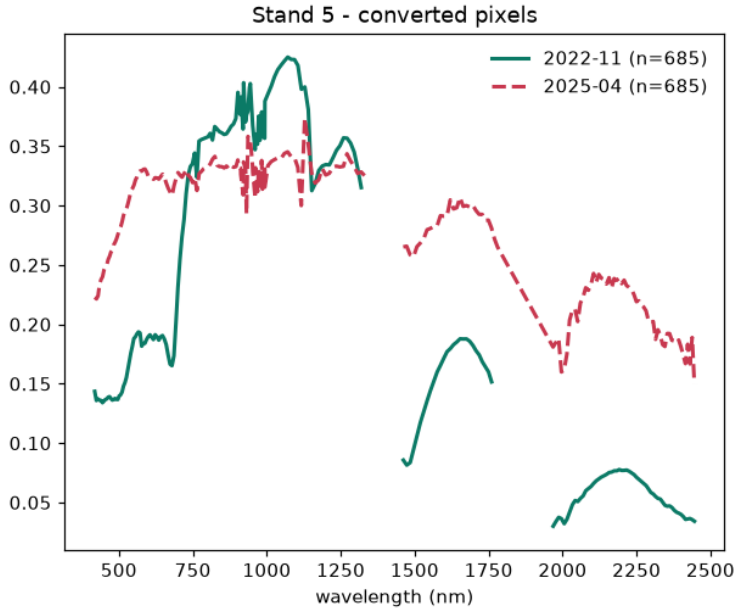
Field carbon plots
Schile et al. 2016 · carbon density
24 plots

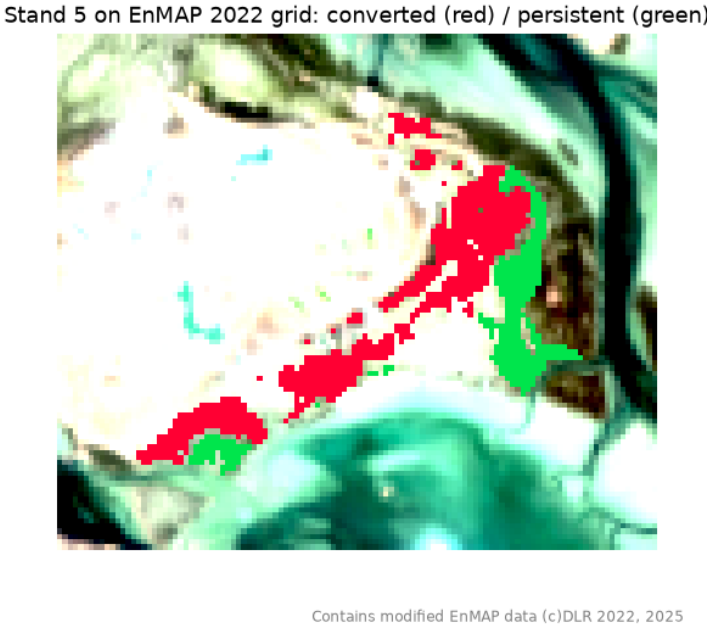
Sub-metre captures
WorldView-2/3, Legion-1 · confirm events
0.3–0.5 m

Quality control: cloud masks · interior pixels only · tide model · second-source checks

A second sensor shows what the mangrove became







Contains modified EnMAP data (c)DLR 2022, 2025

r = 0.84–0.99

agreement with our Sentinel-2 indices on the same dates (25 and 14 stands)

62 vs 26 ha

stand-5 loss: EnMAP pixel count vs our conservative Sentinel-2 flag

Vegetation → mineral

converted pixels take the spectrum of sand and fill: reclamation, not dieback

Honest limit: as a classifier, the full spectrum is no better than Sentinel-2-like bands (F1 0.74 vs 0.72). Contains modified EnMAP data © DLR 2022, 2025.

Sentinel-2 says when. Satellite 813 says what.

Sensor	Bands	Range	Pixel	Revisit	Role in the product
Sentinel-2	13	443–2190 nm	10–20 m	~5 days	WHEN: the alert, every pass
EnMAP (used)	224	420–2450 nm	30 m	~27 days	WHAT: tested on stand 5
Satellite 813 (next)	**~205**	**~400–1700 nm**	**~20 m**	**on demand**	**WHAT, sharper: tasked on flagged cells**

1 · Sentinel-2 alert

stand or 200 m cell below its threshold

→

2 · Task Satellite 813

only over the flagged cells, on demand

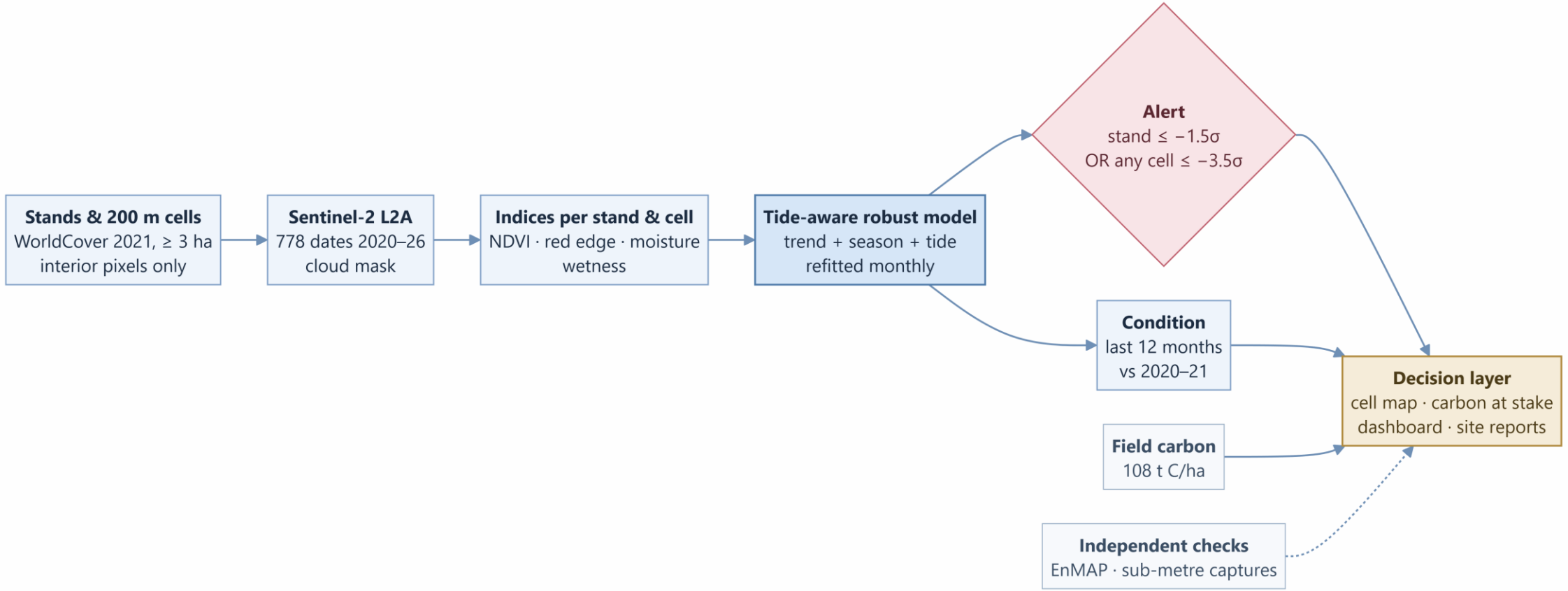
→

3 · Spectral diagnosis

vegetation, or sand and fill: an evidence pack

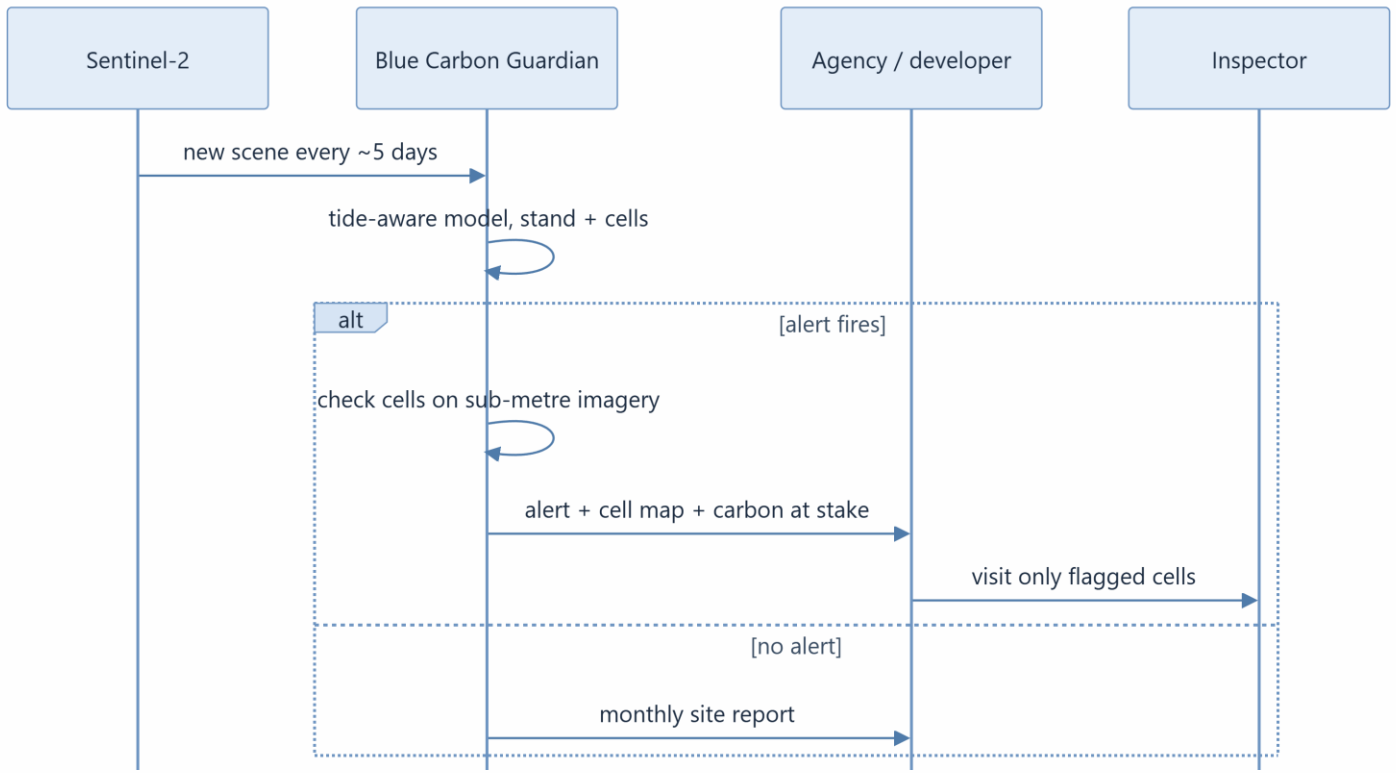
Every index the monitor uses (NDVI, red edge, SWIR moisture at 1610 nm, wetness) sits inside 813's ~400–1700 nm range, at 20 m instead of EnMAP's 30 m. **Simulated with EnMAP's 400–1700 nm bands:** the stand-5 loss still reads vegetation → sand and fill (97 % of converted pixels, 0.4 % of intact ones). Not yet on real 813 data.

From satellite pixels to a decision



Read left to right: Core model · Alert rule · Output to the user · Inputs and checks

One alert, one message, one targeted visit



Every ~5 days

a new Sentinel-2 scene is read in place and run through the tide-aware model, per stand and per 200 m cell

If an alert fires

we check the flagged cells on sub-metre imagery, then send the alert by email or JSON API with the cell map and the carbon at stake

If nothing changes

the user still gets a monthly site report: condition, trend and carbon stock for every stand

AFTER BAHRAIN SPACE AGENCY FEEDBACK

Every flag becomes a case

4 · Urgent **4 · Monitor** **1 · Closed** date · likely cause · confidence · inspector action · open until a person closes it

R-05 · Stand 5 — Urgent

Open since	25 Oct 2022
Likely cause	Coastal development: reclamation with finger canals
Confidence	**high** · Confirmed on dated sub-metre imagery (0.3-0.5 m); stand alert and cells agree.
Inspector action	Inspect the 28 flagged cells (75.3 ha); compare the works boundary with the permit; record the loss for carbon accounting.

75.3 ha to inspect

2.8–6.7 kt C carbon at stake

R-25 · Stand 25 — Monitor

Open since	8 Nov 2024
Likely cause	Unknown: not yet checked on imagery
Confidence	**low** · 1 cell(s) below the 2020-21 baseline; one sensor, not yet checked.
Inspector action	Check the latest sub-metre image of cell 614 before sending anyone.

2.2 ha to inspect

up to 0.2 kt C carbon at stake

R-03 · Stand 3 — Closed

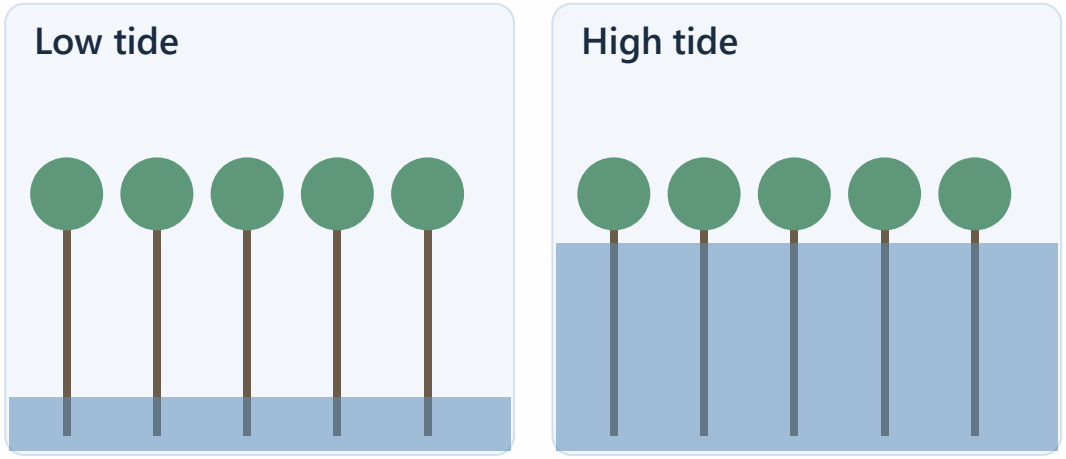
Closed on	26 Oct 2026
Likely cause	Low tide in the image: the canopy was greener than in the previous ten images, so a tide effect, not a loss
Confidence	**checked: no loss** · Checked by a person on the image data.
Inspector action	None. Closed with the reason recorded.

Live on the website (Risks section) and as a data feed: api/v1/risks.json

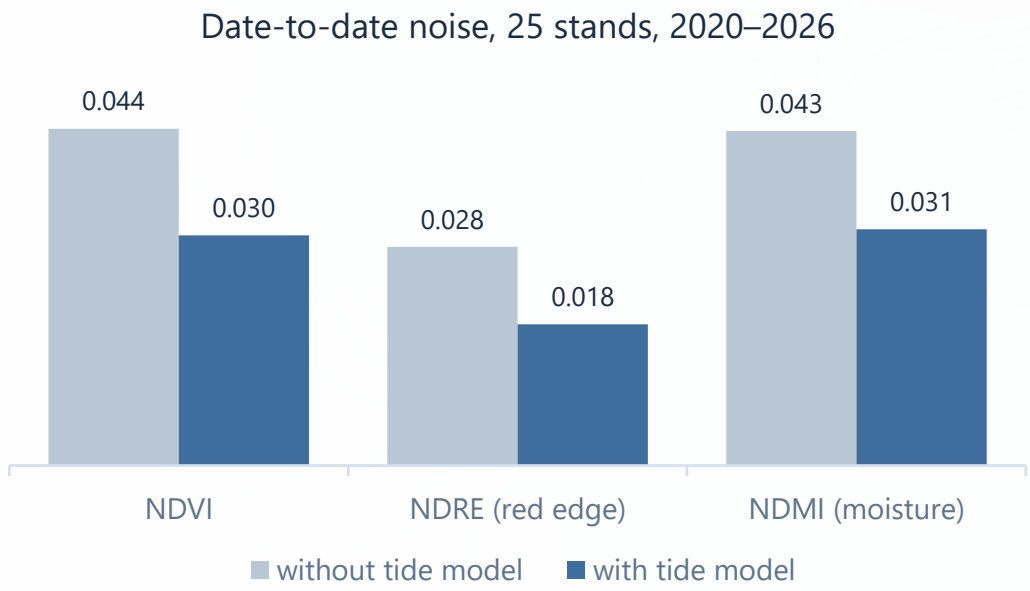
TECHNICAL INNOVATION 1

The tide looks like tree loss — until you correct for it

WHY SIMPLE MONITORS STRUGGLE ON TIDAL FLATS



Same trees, same season. Measured on our stands: noise vs wetness r ≈ −0.9.



−29 to −35 %

less noise on every index: we model trend, season and the tide before flagging anything

1.61 → 0.96

false alarms per stand-year: NDVI alone vs our tide-aware monitor, same alert rule, same 25 stands (40 % fewer)

A stand average hides partial loss. 200 m cells do not.



Real stand-5 data, 2022 → 2026: cells turn red while the stand average barely moves.

Partial loss caught within 60 days (simulated)

73 % stand + cells

25 % stand alone

13 % by chance

8 of 28

cells of stand 9 in severe decline while its average reads “stable”

3 of 5

real conversions found only by the cells (stands 12, 18, 6)

ONE ALERT RULE

stand ≤ −1.5σ OR any cell ≤ −3.5σ

calibrated together: 0.96 false-alarm episodes per stand-year

METHOD IN ONE SLIDE

How the numbers are made

TIDE, FROM THE IMAGE ITSELF

MNDWI = (Green − SWIR) / (Green + SWIR)
flooded share = % of pixels with MNDWI > 0

No tide table: the image shows how much water is under the trees at that exact moment (Sentinel-2 bands B3 and B11).

EXPECTED VALUE, PER STAND AND CELL

index ≈ trend + season + tide
tide = MNDWI, MNDWI², flooded share

Robust regression for NDVI, red edge and NDMI, refitted every month on the past 3 years only. The tide terms cut noise by 29–35 %.

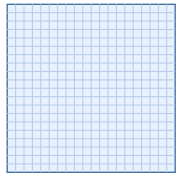
ALERT

z = (observed − expected) / robust σ
alert: stand ≤ −1.5σ OR any cell ≤ −3.5σ

Median of the last 5 images, so one bad image cannot fire it. Thresholds set on unchanged stands: about 1 false alarm per stand per year.

WHY 200-METRE CELLS

200 m = 20 × 20 Sentinel-2 pixels
= 400 pixels = 4 ha



1 cell = 400 px

Enough pixels to average out noise, small enough to send an inspector to 4 ha. 3 of the 5 real losses were found only by cells.

Carbon = area × 108 t C/ha (field plots, Schile et al. 2017). Full equations and sources: README › References and equations.

How we differ from what exists today

	Field surveys	One-off satellite maps	Free loss alerts (Global Mangrove Watch)	Blue Carbon Guardian
How often	campaigns	yearly or less	monthly	**every ~5 days**
Corrects for the tide	n/a	not usually	not documented	**✓ tide model**
Says where inside a stand	✓	—	—	**✓ 200 m cells**
Published false-alarm rate	n/a	n/a	accuracy only	**✓ ≈ 1 per stand-year**
Carbon at stake per alert	—	—	—	**✓ field-based**
Gulf coverage	✓	✓	not confirmed	**✓ Abu Dhabi + Tarut Bay**

Complementary, not competing: Global Mangrove Watch flags loss worldwide; we add the tide model, the 200 m cells, a measured false-alarm rate and the carbon at stake — built and tested for the Gulf.

Five real conversions, confirmed on sub-metre imagery



Stand 5 reclamation with finger canals



Stand 9 canal development, south-east

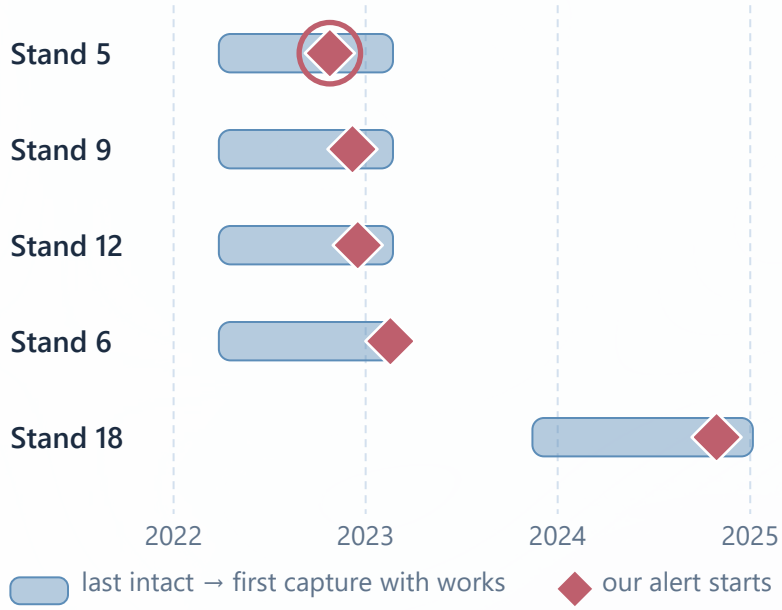


Stand 12 found only by the cells



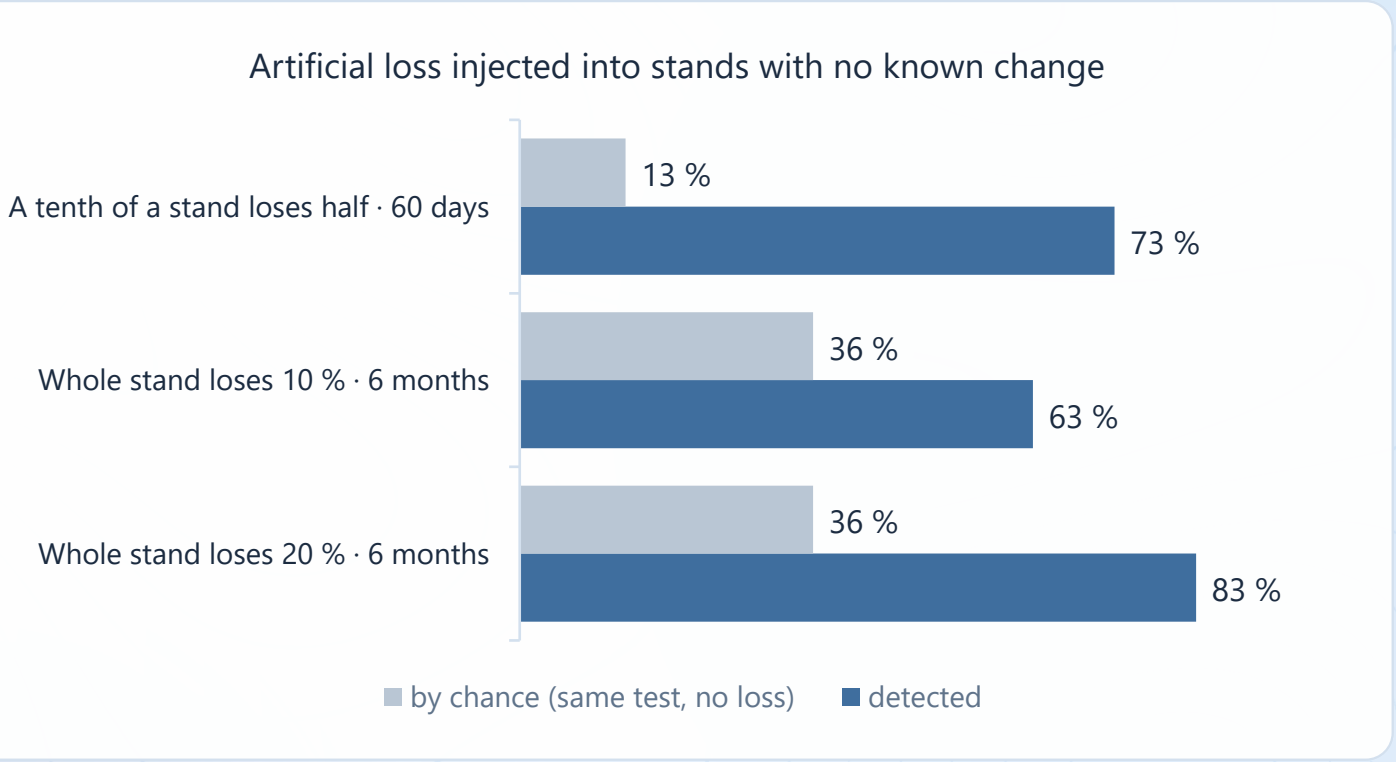
Stand 18 found only by the cells, 2025

WHEN DID THE ALERT START?



Every alert began inside the window. Detection as works happen, not advance warning.

Every rate next to its chance rate



CALIBRATED · 25 STANDS

0.96

false-alarm episodes per stand-year

UNSEEN · 65 STANDS

0.79

per stand-year, 305 stand-years, incl. Saudi Tarut Bay

ENMAP · SAME DATES

r = 0.84–0.99

vs our Sentinel-2 indices; stand-5 loss 62 ha vs our 26 ha

A detection counts only if a new alert starts after the loss begins. ~160–240 trials per row.

What we do not claim: advance warning, field verification, species discrimination, or that hyperspectral classifies better (F1 0.74 vs 0.72).

IMPACT & BUSINESS VIABILITY

Carbon at stake, and who pays to protect it

108 t C/ha

field density near the stands (90 % CI 79–138); 82 % of it is in the soil

247 kt C

pilot stock, 2,281 ha (range 180–316) ≈ 907 kt CO_2e

2.8–6.7 kt C

stock on the 26–62 ha lost at stand 5: what one alert puts on the table

SERVICE TIERS (PRICES ARE PILOT HYPOTHESES)

Monitor

$12k

per site / yr

alert, cell map, condition, carbon, monthly reports

Evidence

$25k

per site / yr

+ sub-metre check of every alert, audit pack

Development watch

$8k

per project / yr

for coastal developers and EIA consultants

Cost to serve ≈ $3.3k / site / yr: compute < $5 (measured), sub-metre checks $500–750, analyst $2k (assumption).

COMPETITION

Global Mangrove Watch sends free monthly loss alerts (Gulf coverage not confirmed). We are complementary: tide-aware, calibrated, stand + cell, carbon and reports.

THE HONEST MARKET

Gulf + Red Sea subscriptions ≈ $0.2–0.4 M / yr. The real prize: a **national MRV contract** or a monitoring layer on **gIQ**. SDG 13 · 14 · 15.

Carbon at stake = lost area × field carbon density: 26–62 ha (our Sentinel-2 and EnMAP estimates) × 108 t C/ha (90 % CI 79–138; Schile et al. 2017, Ecol. Appl. 27:859; data Dryad doi 10.15146/R3K59Z) = 2.8–6.7 kt C; × 3.67 = t CO_2e. Cost to serve = compute < $5 (measured; Azure D8s v5 $0.384/h) + sub-metre checks $500–750 (SkyWatch $19–29/$km^2$) + analyst $2k (assumption). Prices are pilot hypotheses.

Reproducible today, built to scale tomorrow

DOCUMENTATION & EVIDENCE

Public repository, MIT licence
code, notebook, figures, provenance and licences

Re-runs end to end in ~24 s
Python 3.12 · Planetary Computer · pinned requirements

Clean-machine test passed
fresh environment, same numbers, 0 errors

Every number traceable
each figure comes from a script and a written summary

Live website + site reports
anyone can open the monitor and check a stand

DEVELOPMENT POTENTIAL

Proven now
91 stands · 2 countries · 2 Sentinel-2 tiles · 65 unseen stands at 0.79 false alarms / stand-yr · compute < $5 per site per year

Incubation
Satellite 813 in the spectral step · deployment on gIQ · a pilot customer + site records · recovery tracking for planting

Scale
Gulf ≈ 165 km^2 + Red Sea ≈ 200 km^2 of mangrove · sub-metre data for small stands (Bahrain) · national MRV, licensing

Areas: Frontiers in Marine Science 2025. Compute measured; prices assumed.

WHY IT MATTERS IN THE REGION

Big pledges, and a coast that is changing fast



UAE

100 M

mangroves by 2030 · MOCCAE, COP26. First GCC monitoring guide: EAD with IUCN, Feb 2026.



Saudi Arabia

100 M+

mangroves by 2030 · Saudi Green Initiative. Red Sea Global alone: 50 million by 2030.



Bahrain

×4

its mangroves by 2035 · pledged at COP26 (2021). Small stands: they need higher-resolution imagery.



Arab region

≈ 365 km²

of mangrove on the Gulf (≈ 165) and Red Sea (≈ 200) coasts; monitoring still “uneven”.

5 / 25

pilot stands lost mangrove to coastal development in 2022–25, confirmed on sub-metre imagery (our result).

$25–29

per tonne of CO_2e: blue-carbon credit prices 2024–25 (S&P Global Platts). A lost hectare is also lost value.

Tested: Abu Dhabi + Saudi Tarut Bay · same species across the Gulf (Avicennia marina) · Next: Bahrain, sharper imagery

Serves the UN Sustainable Development Goals



Sources: MOCCAE COP26 pledge (2021) · EAD–IUCN GCC mangrove monitoring guide (Feb 2026) · Saudi Green Initiative · Red Sea Global (2023) · Bahrain four-fold by 2035 (GDN, COP26) · Frontiers in Marine Science 2025, doi 10.3389/fmars.2025.1695426 · S&P Global Platts 2024–25

RISK REGISTER · RANKED BY PRIORITY

What to inspect first

Every flag becomes a case with a date, a likely cause, a confidence level and an action for the inspector. A case stays open until a person checks it and closes it with a reason.

4 · Urgent 1 · Monitor 1 · Closed

R-05 · Stand 5

Urgent

Open since	25 Oct 2022
Likely cause	Coastal development: reclamation with finger canals
Confidence	**high** · Confirmed on dated sub-metre imagery (0.3-0.5 m); stand alert and cells agree.
Inspector action	Inspect the 28 flagged cells (75.3 ha); compare the works boundary with the permit; record the loss for carbon accounting.

75.3 ha to inspect

R-09 · Stand 9

Urgent

Open since	7 Dec 2022
Likely cause	Coastal development: canals in the south-east
Confidence	**high** · Confirmed on dated sub-metre imagery (0.3-0.5 m); stand alert and cells agree.
Inspector action	Inspect the 8 flagged cells (21.8 ha); compare the works boundary with the permit; record the loss for carbon accounting.

21.8 ha to inspect

R-12 · Stand 12

Urgent

Open since	17 Dec 2022
Likely cause	Coastal development: canal and reclamation at the southern edge
Confidence	**high** · Confirmed on dated sub-metre imagery (0.3-0.5 m) alert and cells agree.
Inspector action	Inspect the 6 flagged cells ha); compare the works boundary with the permit; the loss for carbon accoun

13.2 ha to inspect

Scan: live website

BAHRAIN SPACE AGENCY

Expert Evaluation and Feedback

وكـــالة البحريـن للفضـاء
BAHRAIN SPACE AGENCY

Specialised assessment and technical feedback

for research that advances Bahrain's environmental goals

OFFICIAL STATEMENT

The Bahrain Space Agency commends Ms. Nahla Nabil for presenting Blue Carbon Guardian, an innovative project using Earth Observation data to monitor regional mangrove ecosystems. The Agency provided specialised assessment and technical feedback, reaffirming its commitment to supporting research and innovation that advance Bahrain's environmental objectives and sustainability goals.

— Bahrain Space Agency, official statement

Meeting and written feedback, 26 Oct 2026 ·
Eng. Roaya Bubshait, Head of Satellite Images & Data Analysis,
with Eng. Aysha Aljawder, Eng. Aysha Almahmeed and Eng. Suad Khalifa

POTENTIAL ADOPTERS IN BAHRAIN · supporting the national pledge to quadruple mangroves by 2035

Bahrain's mangrove stands are smaller than Abu Dhabi's: monitoring them needs higher-resolution imagery than open data (10 m) provides.

Supreme Council for Environment
tracking the national mangrove target

Municipalities & Agriculture
planting and coastal permits

Ministry of Municipalities Affairs and Agriculture

Universities
Earth-observation research

AOU الجامعة العربية المفتوحة Arab Open University البحرين

TEAM BLUE ATHAR · الأثر الأزرق

Every lost mangrove leaves a trace. We find it from space.

Overall value: a working product · five real events confirmed · every number reproducible

Tide-aware	5 of 5	0.79	Open
29–35 % less noise	conversions alerted during the works, 3 only by cells	false alarms / stand-yr on unseen stands	public code, re-runs in ~24 s

Incubation plan (proposed): 30 days: partner, pilot sites, success criteria · 60 days: real Satellite 813 scenes, alerts vs site records · 90 days: pilot on GIQ, auditable reports.

Scan: live website

nahla-nabil.github.io/blue-carbon-guardian · github.com/Nahla-Nabil/blue-carbon-guardian



Thank you

شكراً لدولة الإمارات العربية المتحدة

To the United Arab Emirates for leading the region in protecting its mangroves, and for opening space to Arab youth.

With thanks to: UAE Space Agency · National Space Academy · Space42 · the mentors and the judges



From Bahrain, with gratitude | Nahla Nabil · Team Blue Athar

REFERENCES

Where every number comes from

EXTERNAL SOURCES

UAE: 100 M mangroves by 2030 Ministry of Climate Change and Environment, COP26 (2021); Abu Dhabi Mangrove Initiative, EAD (2022)

Saudi Arabia: 100 M+ by 2030 Saudi Green Initiative (sgi.gov.sa); Red Sea Global, 50 M by 2030 (2023)

Bahrain: ×4 mangroves by 2035 COP26 pledge (2021), GDN Online

GCC mangrove monitoring guide EAD with IUCN Mangrove Specialist Group (Gulf News, 24 Feb 2026); Guidelines for Mangrove Restoration in the UAE (EAD, MOCCAE, Emirates Nature–WWF)

Mangrove areas, "uneven" monitoring Frontiers in Marine Science (2025), doi 10.3389/fmars.2025.1695426

108 t C/ha (90 % CI 79–138) Schile et al. 2017, Ecological Applications 27:859–874, doi 10.1002/eap.1489; field data Dryad doi 10.15146/R3K59Z (CC0)

$25–29 per t CO_2e S&P Global Platts blue-carbon assessments, 2024–25

Cost to serve SkyWatch imagery pricing $19–29/km^2; Azure D8s v5 $0.384/h; analyst time = assumption

Global Mangrove Watch alerts Bunting et al. 2023, Remote Sensing 15:2050

Data Sentinel-2 L2A (ESA Copernicus, via Microsoft Planetary Computer) · EnMAP L2A © DLR 2022, 2025 · ESA WorldCover 2021 (CC BY 4.0) · Esri World Imagery Wayback (Maxar)

OUR OWN RESULTS · FILE IN THE PUBLIC REPOSITORY

Tide model: −29 to −35 % noise analysis/07_summary.md

40 % fewer false alarms (1.61 → 0.96) analysis/35_textbook_baseline.md

73 % vs 13 % by chance analysis/29_cell_trigger.md, 30_backtest_null.md

0.96 calibrated · 0.79 on 65 unseen stands analysis/31_union_alerts.py, 33_scaleout_union_summary.md

Five real events, alert inside image windows analysis/34_vhr_summary.md, 26_substand_summary.md

EnMAP r 0.84–0.99 · 26–62 ha lost analysis/18_enmap_summary.md

Satellite 813 simulation (97 % / 0.4 %) analysis/36_sat813_summary.md

247 kt C · 2.8–6.7 kt C at stake analysis/13_carbon_summary.md (area × field density)

Prices and unit economics docs/business_plan_draft.md (13 numbered sources)

Everything re-runs notebooks/01_blue_carbon_guardian.ipynb · github.com/Nahla-Nabil/blue-carbon-guardian

BACKUP · METHOD

All the equations behind the numbers

1 · Indices (Sentinel-2 bands)

$$NDVI = (B8 - B4) / (B8 + B4)$$
$$NDRE = (B8A - B5) / (B8A + B5)$$
$$NDMI = (B8A - B11) / (B8A + B11)$$

greenness · chlorophyll (red edge) · canopy moisture

2 · Tide, from the image

$$MNDWI = (B3 - B11) / (B3 + B11)$$

flooded share f = % of pixels with MNDWI > 0

median over interior pixels of each stand / cell, every image

3 · Expected value (per stand, cell, index)

$$index(t) = a + b \cdot t + \text{seasonal (1 and 2 cycles a year)} + e_1 \cdot MNDWI + e_2 \cdot MNDWI^2 + e_3 \cdot f + noise$$

Huber robust regression, refitted monthly on [t − 36 m, t − 3 m]

4 · How unusual is today?

$$\sigma = 1.4826 \times \text{median} \mid r - \text{median}(r) \mid$$
$$z = (\text{observed} - \text{expected}) / \sigma$$

state = median of last 5 of mean(z_NDVI, z_NDRE, z_NDMI)

r = residuals of the fit; one bad image cannot fire an alert

5 · Alert

ALERT if stand state ≤ -1.5 OR any cell state ≤ -3.5

thresholds set on stands with no change → 0.96 false-alarm episodes / stand-year

6 · Condition (last 12 months vs 2020–21)

$$\text{change \%} = (\text{now} - \text{base}) / |\text{base}| \times 100$$

severe: NDVI and NDMI ≤ -50 % · decline: ≤ -15 % and $\leq -3\sigma$

on the tide-corrected series; now = median of the last 12 months, base = median of 2020–21

7 · Carbon

stock = area × 108 t C/ha (90 % CI 79–138)

$CO_2e = C \times 44 / 12$ (= 3.67)

stand 5: 26–62 ha × 108 = 2.8–6.7 kt C

density from 24 field plots near the stands (Schile et al. 2017)

8 · Validation

false-alarm rate = alert episodes ÷ stand-years (no change)

detection = % of simulated losses with a NEW alert ≤ 60 days

chance = the same test with no loss → 73 % vs 13 %

simulated loss = a tenth of a stand loses half its canopy

NDVI alone against our monitor, same alert rule

Configuration	Indices	Tide model	False alarms per stand-year	Real conversions caught
A · NDVI only	NDVI	no	1.61	5 of 5
B	NDVI	yes	1.04	5 of 5
C	NDVI + red edge + SWIR moisture	no	1.36	5 of 5
D · ours	NDVI + red edge + SWIR moisture	yes	0.96	5 of 5

40 % fewer false alarms

20 stands with no known change, 93.9 stand-years: about 16 fewer unnecessary inspections a year for 25 stands.

The tide model is most of the gain

NDVI alone 1.61 → with tide terms 1.04; the extra indices bring it to 0.96.

Honest limit

Thresholds calibrated for our monitor; all four catch the five conversions. The difference is the false-alarm cost, not detection.

Where it breaks and what we do not claim

Detection, not early warning

Alerts fire as works happen. We never claim to see stress before it is visible.

10 m pixels

Patches under ~0.6 ha and young plantings are missed, Bahrain's small stands included.

Imagery, not site visits

Conversions confirmed on dated sub-metre captures, not yet on permits or field records.

Simulated losses

Canopy-cover loss, not leaf stress; every rate shown next to its chance rate.

Hyperspectral

EnMAP confirms our indices but was not a better classifier (F1 0.74 vs 0.72).

Carbon

Area × field density from four sites (2013–14): a stock with a range, never a sequestration rate.

Rule we kept: no detection rate without its chance rate; no claim of advance warning, field verification or species discrimination.